

1.Applications: The Role of the Internet of Things for Increased Autonomy and Agility in Collaborative Production Environments.

Introduction:

- Market is continuous to change by innovation.
- The rate of market changes grew steadily over the previous decades, especially due to the improvement as well as the development of existing and new information and Communication technologies (ICT).
- Potential technologies that can confer this behaviour to collaborative production environments are the Internet of Things and autonomous objects
- The future Internet of Things will use protocols and algorithms which will be based on those we use in the Internet today; it will just extend the capabilities of a more extensive machine-to-machine and human-to-machine communication, resulting in a higher number of specialised communication participants within the Internet.
- Autonomous objects are objects which are equipped with intelligence (small central processing units (CPUs) and algorithms) to be capable of making contextual routing decisions or handling activities. Both, the Internet of Things concepts and the autonomous objects, are complementary.
- The Internet of Things acts as an infrastructure and helps to realise the new systemic characteristics of autonomy and agility by providing an object-oriented information architecture for precise real-time data and ubiquitous Internet access.
- When applying the described concepts consequently, several existing paradigms of production environments are affected. Paradigms like "Just inTime" are no longer applicable, for instance. Most of the existing paradigms have in common that they require deterministic environments.
- Introducing a high degree of freedom leads to non deterministic environments and, as a result, state-of-the-art methods have to be extended.
- This contribution investigates the suitability and cooperation of the Internet of Things and autonomous objects for autonomic and agile processes in collaborative production environments.

Emerging Challenges of Networked Enterprises

- Basic challenge of enterprises challenge their rivals through product innovations, strategic partnerships, procedural efficiency, pricing policies, acquisitions or in exploring new market opportunities more extensive an enterprise network is, the more complex are control, synchronisation, fault recovery and reorganisation of the overall process flow.
- Data Communication
- Collaborative Research Centre (CRC)
- Globalisation has created an international competition of the cheapest production sites as well as very efficient global logistics providers so that distances do not play the same important role as before.
- For the purpose of agile and autonomous processes, the enterprises have to integrate high density informational and control networks, which provide extensive real-time data and enable fine-grained controlling and objective specification for management. This feature can be served by the Internet of Things

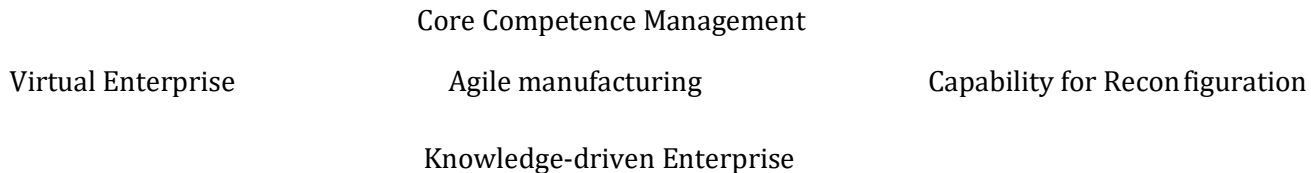
Fundamental concepts of Autonomy and Agility:

Agility

"Agility in manufacturing involves being able to respond quickly and effectively to the current configuration of market demand, and also to be proactive in developing and retaining markets in the face of extensive competitive forces."

"Agility relates to the interface between the company and the market. Essentially it is a set of abilities for meeting widely varied customer requirements in terms of price, specification, quality, quantity and delivery."

Core Concepts of Agile Manufacturing



Autonomous Control

"Autonomous Control describes processes of decentralized decision-making in heterarchical structures. It presumes interacting elements in non-deterministic systems, which possess the capability and possibility to render decisions independently. The objective of Autonomous Control is the achievement of increased robustness and positive emergence of the total system due to distributed and flexible coping with dynamics and complexity."

Characteristics:

Heterarchy

Autonomous objects need exchange with their environment

- Identification, Communication and Cooperation, Sensors, Storage, Actuating elements, User Interface

Objectives:

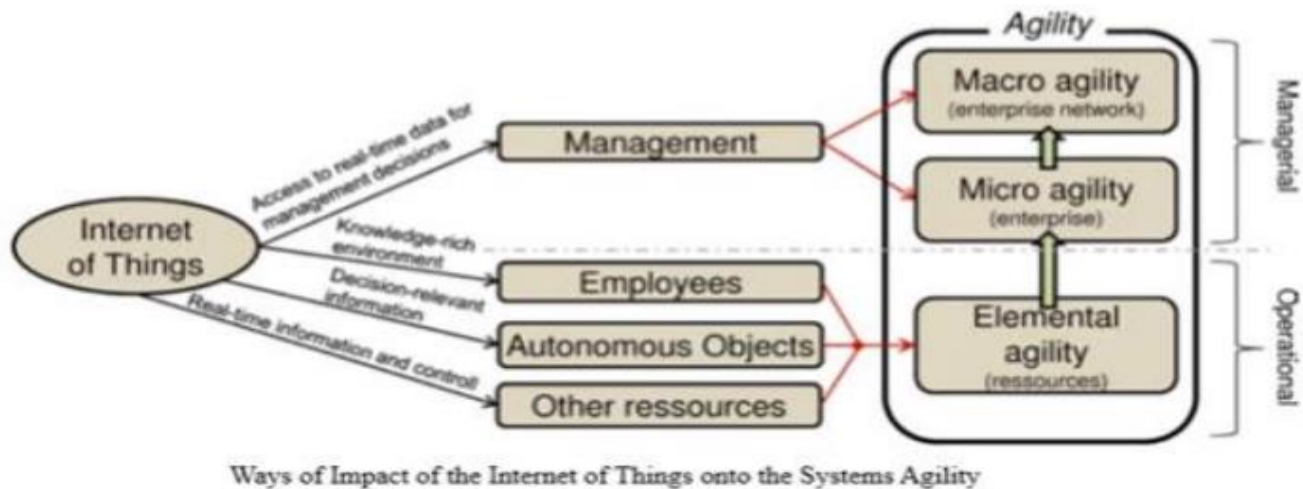
The objective of increased robustness is based on the assumption that autonomous objects can react much faster to unforeseen events than higher planning and controlling instances.

Positive emergence means that the sum of the individual and context dependent decisions, which are made by autonomous objects, gain a better achievement of the total system objectives than it can be explained by the behaviour of every single element

Degree of Autonomous Control :



Enabling Autonomy and Agility by the Internet of Things



Application Field: Automotive Tail-lights- Intelligent Product

- The high complexity of logistics networks makes it more and more difficult to meet the demands of logistics. Having the right product at the right time at the right place - this is becoming very challenging with conventional planning and control methods
- Thus, aspects such as agility, flexibility, proactivity and adaptability are in the focal point of the current research. This is done by applying concepts of decentralisation and autonomy on the logistics decision-making process.
- As a future autonomous logistics object, the Intelligent Product is being introduced. The Intelligent Product will be presented in a production logistics scenario and is capable of acting autonomously through an assembly process.
- This assembly process is part of a production scenario designed to investigate the applicability in the domain of production logistics. The scenario illustrates an autonomous assembly system for an automotive tail-light. The whole equipment is tailored to communicate over the Internet, and, as a consequence, we can imagine extending this scenario in order to consider more than one location of the supply chain.

Assembly Scenario:

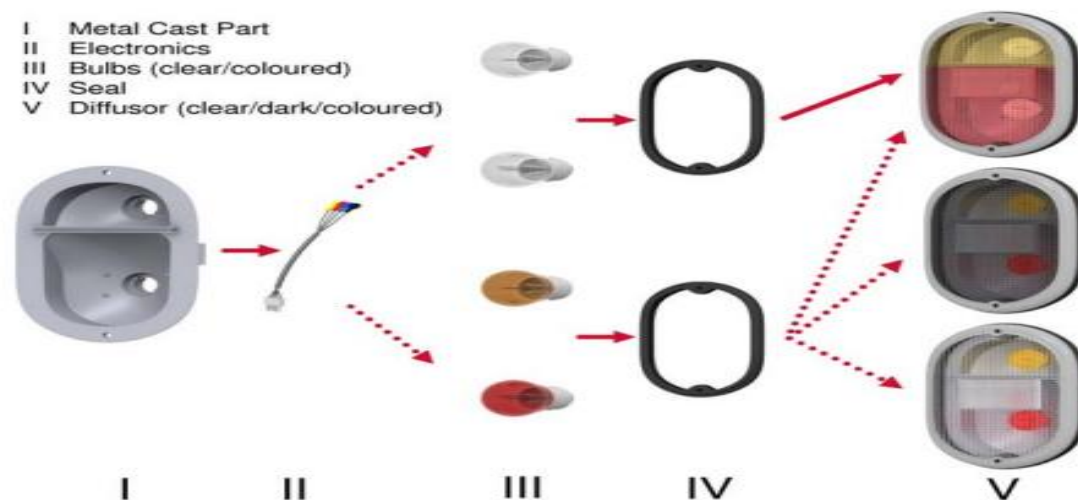


Fig. 8.5 Assembly Process of the Tail-light

Layout

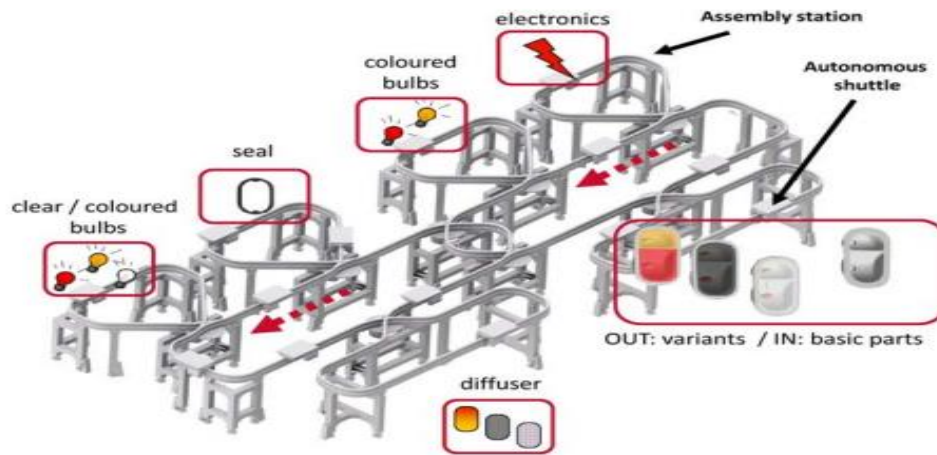


Fig. 8.6 Assembly/Production Scenario (Morales Kluge and Pille 2010)

The System



Technological Prerequisites

- Hardware Abstraction Layer
- Metal Cast RFID
- Multi-Agent-System
- Decision Algorithms

Challenges by Developing the Internet of Things

Authenticity, Encryption and Integrity of Data

- Authentication
 - Legal Safety for Data Protection
 - Scalability
 - Billing and Business Model
- Data Management and Synchronisation
 - Human-to-machine Communication
 - Technological Improvements

2.Resource Management in IoT:

Clustering:

- Clustering is a popular method of organising wireless network topologies, in which a few nodes, the cluster heads (CH),are elected as representatives to route the traffic originated in the entire network.

- Clustering used in Wireless Sensor Networks (WSN) and Mobile Ad-hoc NETWORKS (MANET).•MANET:meshed networks without central authorities.

- MANETs are generally built to handle objects in dynamic environments all nodes act as routers with their own routing tables. clustering approaches of WSN:hierarchical- star,Mesh topology WSN are traditionally used to cluster more or less static nodes We would first like to compare those ad-hoc wireless clustering protocols that consider both mobility and energy-efficiency.

- The studied properties are the following:

- 1.Type:**If the protocol is specifically for WSNs or for more general MANETs.

- 2.Controlled variable CH period:** CHs may be elected for periodic oraperiodic time intervals.

- 3.CH election according to node conditions:** If a node is elected according to its own conditions.

- 4.Synchronisation:** If the nodes need synchronisation for either electing the CH or operating inside the cluster

- 5.Global cluster information:** If the cluster nodes need to store information about all the cluster members in order to perform the CH election or to operate. Global information implies poor scalability with the number of network nodes.

- 6.Multi-hop routing Multi-hop:** routing mechanisms are beneficial because they can route communication packets between two nodes that are not directly connected5

- 7. CH election complexity:** lower the complexity, the more efficient is the proposed algorithm.

Software Agents

- Agent Based Systems are an evolving software paradigm that strives to create software that can possess human characteristics, such as autonomy, adaptability, sociality, judiciousness, mobility and reactivity .

- Intelligent agents are software programs that continuously perform following functions:

- 1.perception of dynamic conditions in the environment;

- 2.reasoning to interpret

- 3.perceptions,

- 4.solve problems,

- 5.draw inferences, and determine actions

- software entity to be named an agent, it should maintain thefollowing properties .

- 1.the environment consists of other agents, the agent needs to have“social-ability

- 2.goal-directed behavior

- 3.Agents might also “learn” to improve their behaviour using feedback from its performance

Data Synchronization:

Data synchronization is the ongoing process of synchronizing data between two or more devices and updating changes automatically between them to maintain consistency within systems.

While the sheer quantity of data afforded by the cloud presents challenges, it also provides the perfect solution for big data. Today's data solutions offer quick and easy tools to bypass monotonous tasks, resulting in data in harmony throughout the system.

Data synchronization ensures accurate, secure, compliant data and successful team and customer experiences. It assures congruence between each source of data and its different endpoints. As data comes in, it is cleaned, checked for errors, duplication, and consistency before being put to use. Local synchronization involves devices and computers that are next to each other, while remote synchronization takes place over a mobile network.

Data must always be consistent throughout the data record. If data is modified in any way, changes must upgrade through every system in real-time to avoid mistakes, prevent privacy breaches, and ensure that the most up-to-date data is the only information available. Data synchronization ensures that all records are consistent, all the time.

Data synchronization: The key to trusted data

The importance of data synchronization grows in step with increased accessibility to cloud-based data as well as access to mobile devices. Mobile devices use data for basic operation as well as personal information for apps, websites, and email. Updates to information generated by the user as well as the end target must be constant and secure. This synchronization process requires clean, consistent data for product and service competence, but also for data governance issues like security and regulatory compliance.

Data conflicts can result in errors and low data quality, which consequently leads to a lack of trusted data down the line. With data synchronization properly implemented throughout a system, a business will see performance improvement in many areas, including:

- Logistics and transportation
- Sales team productivity
- Order management
- Invoice accuracy
- Business systems
- Cost efficiency
- Reputation management

Data accessibility and error resolution afford time savings, allowing emphasis to be put on important business development processes like marketing, new product development, and strategic decision-making. Virtually everyone benefits from clean, synced data.

1. Customers receive product information and service that meets their specific needs
2. Business users can interact with all department members using up-to-date information, in real-time, even globally
3. Executives receive the latest data when making important strategy decisions
4. Stockholders can easily stay on top of their business interests
5. Manufacturers access the most recent updates or changes for accurate design and production
6. Distributors have access to the most recent product and marketing information

3.Applications:

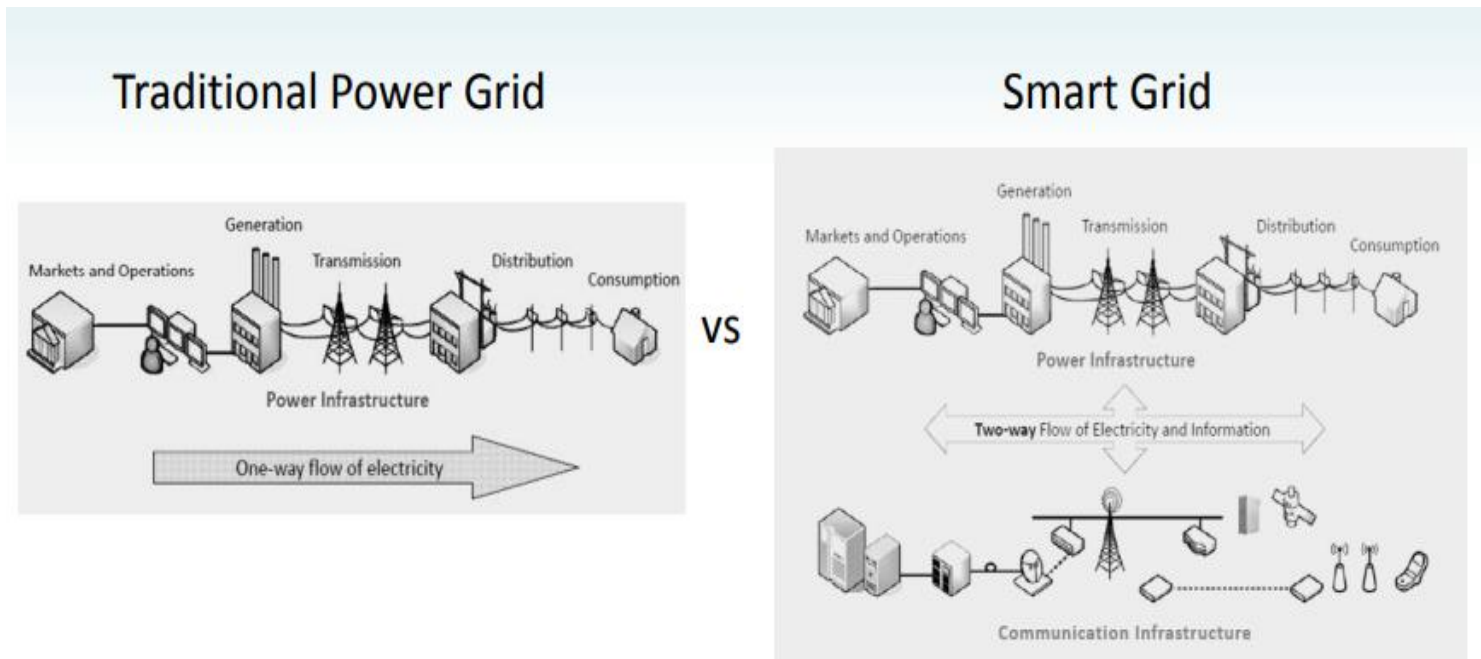
The **Internet of Things (IoT)** provides the ability to interconnect computing devices, mechanical machines, objects, animals or unique identifiers and people to transfer data across a network without the need for human-to-human or human-to-computer is a system of conversation. **IoT applications** bring a lot of value in our lives. The Internet of Things provides objects, **computing devices**, or **unique identifiers** and people's ability to transfer data across a network without the **human-to-human** or **human-to-computer interaction**.

4.Smart Grid:

Internet of Things is now the talk of every city. It witnessed a tremendous growth in every sector. From connected medical devices, connected cars, roads to smart grids for energy management. When we talk about energy management, smart grid is the first thing to appear.

Internet of Things can play a significant role in developing smart grids which eventually leads to energy saving. IoT enabled smart grids are more about the energy efficiency and the management of energy consumption at the lowest cost.

A Smart Grid is an electricity network that can intelligently integrate the actions of all users connected to it – generators, consumers and those that do both – in order to efficiently deliver sustainable, economic and secure electricity supplies.



Smart Grid Components

- Smart Meter
- Phasor Measurement
- Information Transfer
- Distributed Generation

Smart Grid Architecture

1.Conceptual model

Smart Grid is a large "System of Systems". According to Smart Grid Interoperability Standards Roadmap proposed by NIST the American National Institute of Standards and Technology, the conceptual architecture for smart grid is composed of seven big domains

- Bulk Generation
- Transmission
- Distribution
- Customers
- Operations
- Markets
- Service Providers

2. Electrical network

Production Domain composed of a mixture of nuclear, solar, coal, wind or hydro power plant Transmission Domain managed by huge number of network operating centers and substations, a large number of power lines deliver the electricity to distribution domain Distribution Domain sum of complex networks topologies delivers electrical power to residential areas, rural farms, metropolitan areas, and industrial areas for consumption

3. Communication network

Communication networks for smart grid systems use a big range of communication technologies from wired, wireless and hybrid networks technologies.

- Home Area Network (HAN)
- Neighborhood Area Network (NAN)
- Field Area Network (FAN)
- Wide Area Network (WAN)

5. Electrical Vehicle Charging:

With the growing concerns on energy conservation and the escalating emission of carbon dioxide, electric vehicles such as V2Gs, also known as vehicle to grids, have become prevalent all over the world owing to its zero-emission, low-noise, low maintenance and shrinking operation cost.

It is expected that the global electric car stock may reach 40-70 million by 2025. (Source: IEA analysis based on EVI country submissions, complemented by EAFO 2017a).

However, a large quantity of decentralized EV charging stations would be hard to remotely manage as well as run into the following challenges:

- A weak point, first of all, lies in its capability of data transmission. Many stakeholders will be delayed to receive the field data of charging and metering.
- The full suite of services are offered by various providers for which it would be hard for troubleshooting when any issue occurs.
- Most charging points are deployed in challenging environment where it would be critical to approach.
- The scattered charging stations with complex structure will bring a great difficulties to onsite technical personnel for the devices maintenance and management.
- Redundant link support, sometimes however, is also in the list of technical requirements.

In view of the above issues, Ursalink, a united team of qualified professionals that have over 12- year experience in the field of communications, not only provides an easy-setup, operational efficiency, and cost optimization plan for EV charging station, but makes it possible to realize remote control with the wireless 3G/4G design as well.

The solution comes with EV, EV charger, Ursalink 3G/4G cellular router, and cloud management software from EV charger manufacturer or a third party

Wherein, the Ursalink industrial cellular router plays a crucial role of providing a high-speed and stable networking and transparent link for the charging and metering data transmission so as to analyze and monitor the devices with prompt response, by which leads to a desired outcome of cost reduction.



Benefit & Features

1. Ursalink router helps connect to the Internet via wireless 4G/3G/GPRS technology for the charging point placed in an environment where it's too difficult or expensive to deploy optical network
2. Automated fail-over/fail-back between Ethernet and cellular (dual SIMs) feature ensures all data to be sent to the management software timely
3. Multiple options of VPN (IPsec, OpenVPN, DMVPN, GRE, L2TP, PPTP) with different encryption methods ensure the secured and stable communication
4. Featuring 2 or 5 Gigabit Ethernet ports, an Ursalink cellular router is all you need for each charging station and gets you into light transmission of large data from multiple EV chargers
5. Offer serial ports (RS232 & RS485) and I/O interface for receiving and transferring EV chargers' voltage, current, power distribution and other operating parameters
6. SMS alarm & mail alarm system provide ability for engineers to take corresponding actions in no time by sending them the alerts of those out-of-compliance equipment
7. Embedded Ursalink SDK (Python 2.7/C) for secondary development in SSD/Micro SD card
8. Featuring industrial design with IP30 protection class to function properly in the most demanding environment, from cool to crucial.

6.Case studies:

A case study can be defined as an intensive study about a person, a group of people or a unit, which is aimed to generalize over several units.

7.Sensor body-area-network

A sensor network is a group of sensors where each sensor monitors data in a different location and sends that data to a central location for storage, viewing, and analysis.

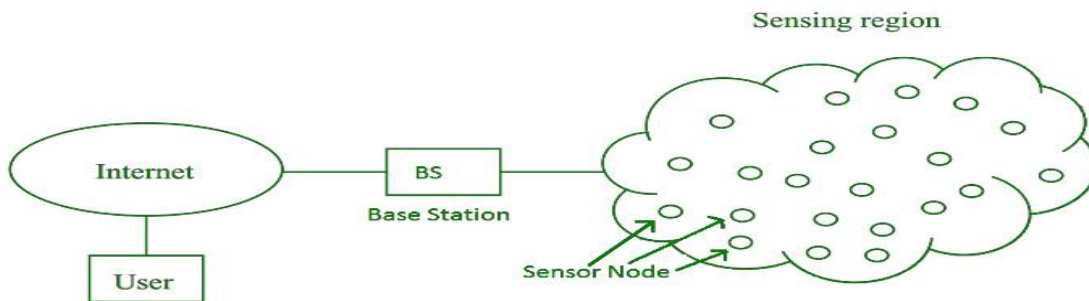
Wireless Sensor Network (WSN):

Wireless Sensor Network (WSN) is an infrastructure-less wireless network that is deployed in a large number of wireless sensors in an ad-hoc manner that is used to monitor the system, physical or environmental conditions.

Sensor nodes are used in WSN with the onboard processor that manages and monitors the environment in a particular area. They are connected to the Base Station which acts as a processing unit in the WSN System. Base Station in a WSN System is connected through the Internet to share data. WSN can be used for processing, analysis, storage, and mining of the data.

Applications of WSN:

1. Internet of Things (IoT)
2. Surveillance and Monitoring for security, threat detection
3. Environmental temperature, humidity, and air pressure
4. Noise Level of the surrounding
5. Medical applications like patient monitoring
6. Agriculture
7. Landslide Detection



Components of WSN:

1. **Sensors:**
Sensors in WSN are used to capture the environmental variables and which is used for data acquisition. Sensor signals are converted into electrical signals.
2. **Radio Nodes:**
It is used to receive the data produced by the Sensors and sends it to the WLAN access point. It consists of a microcontroller, transceiver, external memory, and power source.
3. **WLAN Access Point:**
It receives the data which is sent by the Radio nodes wirelessly, generally through the internet.
4. **Evaluation Software:**
The data received by the WLAN Access Point is processed by a software called as Evaluation Software for presenting the report to the users for further processing of the data which can be used for processing, analysis, storage, and mining of the data.

Body Area Network (BAN):

Body Area Network (BAN) technology uses small, low power wireless devices that can be carried or embedded inside or on the body.

In today's world, wireless communication has a major application in sharing of information anywhere and at anytime. We can use wireless networks in the form of WLAN or Wi-Fi in various fields such as education, healthcare, and industrial sector. As the technology is growing, the demands of users as well as the demand of ubiquitous networking is increasing. WBAN(Wireless Body Area Network) allows the user to move another without having the restriction of a cable for sharing information.

The term 'Wireless Body Area Network' was coined in 2001 by Van Dam. It basically is a network containing sensor nodes that are attached to the human body, used to measure the bio signals(heart rate, blood pressure, brain signals, etc.) of humans. It has majority of applications in medical sector.

The communication in body sensor networks is of 2 types:

- **In-body communication**
- **On-body communication**

In-body communication is the communication between sensor nodes that are implanted inside human body. The **MICS** (*Medical Implant Communication System*) communication can be used only for in-body communication.

On-body communication occurs between wearable devices which consist of sensor nodes. The **ISM** (*Industrial Scientific and Medical*) band and **UWB** (Ultra-wideband) communication can be used only for on-body communication.

WBAN requirements:

- Low power consumption
- Interoperability
- Self-healing
- Security
- Low latency

WBAN architecture:

The network architecture is divided into 4 sections-

1. WBAN Part –

It contains several number of cheap and low-power sensor nodes, which can be used for continuous monitoring of heart rate, ECG, blood pressure, etc. of a person. Being wireless in nature, this does not restrict the mobility of the person for continuous evaluation. Hence, WBAN is used in healthcare systems for patients monitoring.

2. CCU (Central Control Unit) –

All sensor nodes provide their outputs to a central coordination node present in the CCU. CCU receives the signals from nodes and transmits it to the next section for monitoring the human body.

3. WBAN communication –

Receives information from CCU and acts as gateway to transfer information to the destination. For ex. mobile node is a gateway to remote station to send message to cellular network using GSM/3G/4G.

4. Control center –

It is responsible for storing the information of user which can be used in the future or for monitoring purpose. It consists of end node devices like mobile phones(for messaging), computer systems (for monitoring), and server(for storing information in database).

WBAN applications:

These are various applications:

1. Medical Applications:

- **Remote healthcare monitoring** – Sensors are put on patient's body to monitor heart rate, blood pressure and ECG.
- **Telemedicine** – Provides healthcare services over a long distance with the help of IT and communication.

2. Non-medical Applications:

- **Sports** – Sensors can be used to measure navigation, timer, distance, pulse rate, and body temperature.
- **Military** – Can be used for communication between soldiers and sending information about attacking, retreating or running to their base commander.
- **Lifestyle and entertainment** – Wireless music player and making video calls.

WBAN is legal, affordable, and user-friendly. It is an emerging technology and is expected to have a big impact on the society.

8.Control of a smart home:

A smart home is a residence that uses internet-connected devices to enable the remote monitoring and management of appliances and systems, such as lighting and heating. A smart home's devices are connected with each other and can be accessed through one central point—a smartphone, tablet, laptop, or game console.

Door locks, televisions, thermostats, home monitors, cameras, lights, and even appliances such as the refrigerator can be controlled through one home automation system. The system is installed on a mobile or other networked device, and the user can create time schedules for certain changes to take effect.

Smart home appliances come with self-learning skills so they can learn the homeowner's schedules and make adjustments as needed. Smart homes enabled with lighting control allow homeowners to reduce electricity use and benefit from energy-related cost savings.

Some home automation systems alert the homeowner if any motion is detected in the home when they're away, while others can call the authorities—police or the fire department—in case of imminent situations.

A **smart home system** can be something that makes our life quite easy. Starting from energy management where the power controls system in the AC appliances where we use the thermostat, all this is managed to cut down the power consumption that's taking place.

A door management system, security management system, water management system are the part of this as well. Still, these are vital things that stand out in the smart home system. The limitation of IoT in smart home application stops where our imagination stops. Anything that we wish to automate or want to make our life easier can be a part of smart home, a smartphone system as well.

Smart Home

